Measuring Teacher Attrition with Labour Force Survey Data

The Case of the United States

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Why this project?

Motivation

- Understand why teachers leave the profession.
- Is there a real policy margin to soften teacher shortages by retaining teachers, or bringing them back?

Problem

- Official microdata on teachers are not publicly available for many countries.
- We start with the United States: the country with the most information available.

▸ official sources

Contribution

Use the Current Population Survey to

A. reproduce and extend the official statistics on teachers leaving the profession;

B. having achieved A, study in detail *how many* leave, *where* they go, and *why*.

0

Methodological discussion

The Current Population Survey

- The monthly household survey of the United States, run by the Census Bureau for the Bureau of Labor Statistics. Nationally representative of the civilian non-institutional population.
- We pool 28 consecutive March supplements, 1998–2025: **5.0 million** person-year records.
- **104,545** of them are teachers: about 3,700 per year.

unemployment | interviews | migration | identity

The March supplement (ASEC)

- Run by the Census Bureau each March since 1948: the source of the official annual income, poverty and health-insurance statistics.
- About 90,000 households: the March sample plus oversamples interviewed in February and April.
- Every adult reports on the *previous calendar year*: longest job held, weeks worked, earnings, benefits.

Identifying a teacher: the census occupation codes

	1990 classification surveys 1998–2002	2002/2010 classification surveys 2003–2019	2018 classification surveys 2020–2025
Pre-K and kindergarten	155	2300	2300
Elementary (and middle)	156	2310	2310
Secondary	157	2320	2320
Special education	158	2330	2330
Other teachers and instructors	159	2340	2360

The K–12 block maps one-to-one across the three regimes: the *group* is constant over time even where single codes are not.

▸ traceability


interviewed 4 months
8 months out
interviewed 4 months
1
2
3
4
5
6
7
8
Mar
Jun
Mar
Jun
year 1
year 2
the same interview, twelve months later

Two ways to identify a leaver

A. Follow the individual

All months. Links require matching:

household identifier
dwelling identifier
person line number

validated on sex, race, age +0–2

B. Ask about last year

The March supplement, retrospectively.

Leavers are the ones who taught as last year's longest job and are not teaching this March.

Disadvantages of each option

A. Following individuals

1. Identity is not certain: validation can admit noisy people.
2. Movers are lost: $\sim$25% of addresses are not found a year later.
3. Normal household attrition on top.
4. Strong seasonality in teaching.
5. Many ways to define the rate (next slide).

▸ linkage loss ▸ definition sensitivity ▸ validation

B. Retrospective March

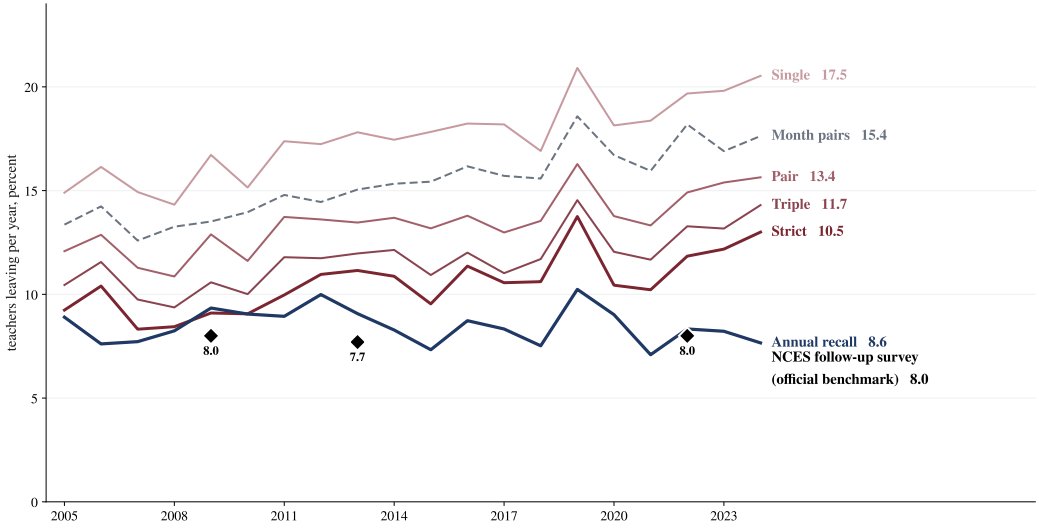
1. Recall bias.
2. Less sample: one observation per person per year ($\sim$3,700 teachers) against $\sim$11,000 monthly teacher observations.
3. Only March-to-March transitions; within-year spells are smoothed away.

Six ways to define leaving, simplest to strictest

Unit: the individual. Four interviews in year 1, four in year 2.

	A teacher:	A leaver:
Single	seen teaching in $\geq$ 1 of the 4 year-1 interviews	seen teaching in none of the 4 year-2 interviews
Pair	seen teaching in $\geq$ 2 of the 4	seen teaching in none of the 4
Triple	seen teaching in $\geq$ 3 of the 4	seen teaching in none of the 4
Strict	seen teaching in all 4	seen teaching in none of the 4
Month pairs	each teaching month, on its own	not seen teaching twelve months later
Annual recall	teaching was last year's longest job	not teaching at the March interview

The six, on one axis

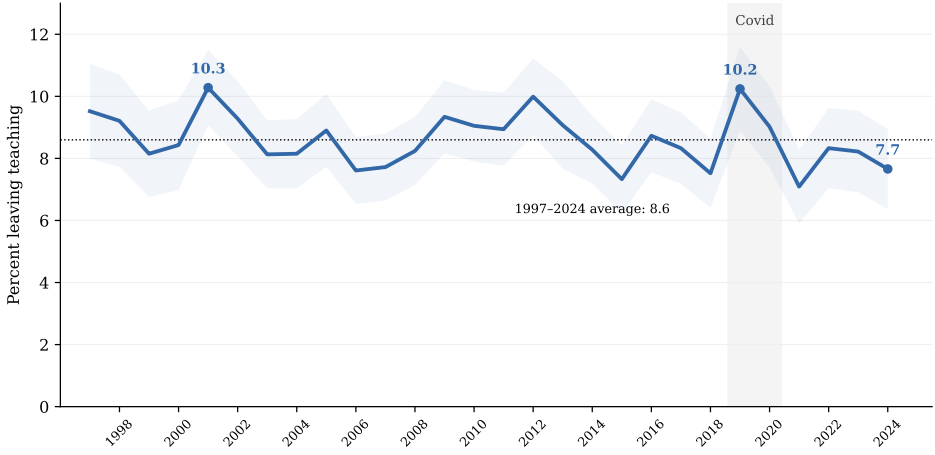


NCES: the National Center for Education Statistics; its Teacher Follow-up Survey is the official benchmark.

1

Leaving rate: how many leave

Leaving teaching, 1997–2024



Message. Around 8 of every 100 teachers leave each year, and the rate has been stable for 25 years. Attrition is a permanent flow, not a recent crisis.

▸ public vs private

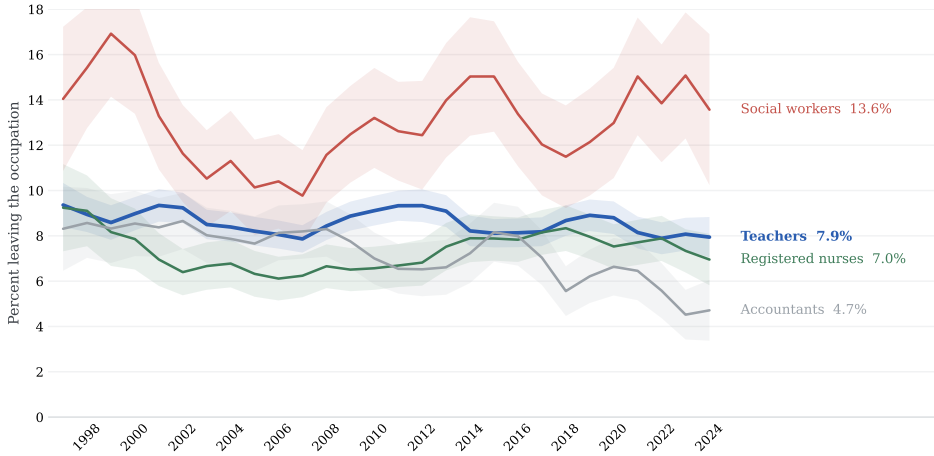
Comparable professions: who we compare teachers with

	Teachers	Nurses	Social workers	Accountants
Female (%)	76.0	88.2	82.9	57.0
Average age (years)	44.4	43.1	43.4	44.1
Master's or higher (%)	51.9	25.9	50.8	31.3
Public sector (%)	70.2	12.3	43.1	11.0
Weeks worked	46.6	49.3	49.6	49.5
Median annual wage ($)	56,000	80,000	57,000	77,000
Observations	28,932	13,107	3,407	6,897

College graduates, pooled 2015–2024, weighted; wages from 2020–2024.

Message. Teaching shares its age and gender profile with nursing and social work. What sets it apart is pay and the public sector.

Leaving the occupation, by profession



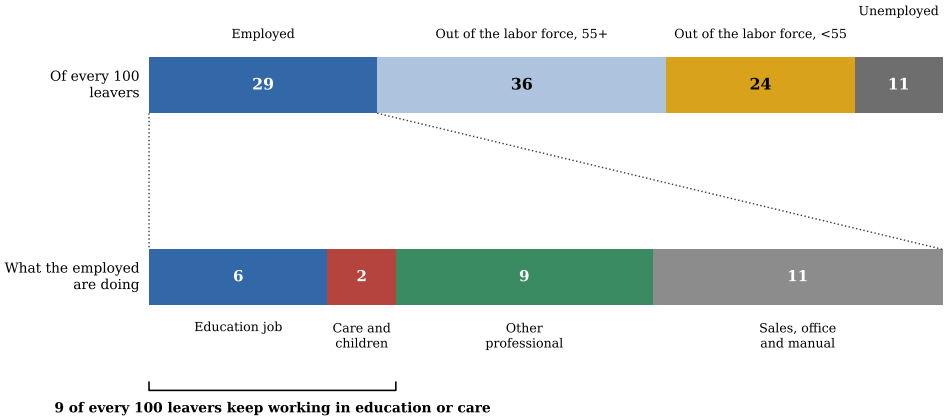
Message. Teachers leave at about the same rate as comparable professions. *How many* leave is not what makes teaching special; *who* leaves and *where they go* is (next block).

▸ more professions

2

Composition: where they go

Of every 100 leavers (average, 2020–2024)



Message. A third find another job; a third retire; the rest split between early labor-force exit and unemployment.

▸ vs NCES follow-up

Where the employed leavers go

red: fewer than half of US workers in the occupation hold a college degree (ASEC 2021–2025)

EDUCATION, NON-TEACHING — 23%	
8.1	Postsecondary teachers
4.3	Tutors
4.2	Teaching assistants
3.1	Education administrators
1.5	Librarians
1.3	Other instruction and library wo...
0.2	Archivists, curators, and museum...
CARE — 9%	
1.8	Educational and career counselors
1.2	Social and human service assista...
1.1	Mental health and substance abus...
1.1	Social workers, all other
0.8	Childcare workers
0.6	Mental health counselors
0.5	Counselors, all other
0.5	Substance abuse and behavioral d...
0.4	Other community and social servi...
0.4	Probation officers and correctio...

PROFESSIONAL & MANAGERIAL — 32%	
9.7	Managers, all other
1.9	Coaches and scouts
1.7	Architects
1.5	General and operations managers
1.3	Business operations specialists...
1.2	Other mathematical science occup...
1.1	Writers and authors
0.9	Broadcast, sound, and lighting t...
0.8	Chemical technicians
0.7	Social and community service man...
0.6	Property, real estate, and commu...
0.6	Photographers
0.6	Actors
0.6	Therapists, all other
0.5	Lodging managers
0.5	Construction managers
0.5	Miscellaneous health technologis...
0.4	Other psychologists
0.4	Other engineering technologists...
0.4	Management analysts
0.4	Market research analysts and mar...
0.4	Human resources workers
0.4	Veterinarians

0.4	Chief executives
0.4	Dietitians and nutritionists
0.4	Information security analysts
0.3	Claims adjusters, appraisers, ex...
0.3	Physical therapists
0.3	Food service managers
0.3	School psychologists
0.3	Editors
0.3	Transportation, storage, and dis...
0.2	Meeting, convention, and event p...
0.2	Training and development managers
0.2	Accountants and auditors
0.2	Other healthcare practitioners a...
0.2	Graphic designers
0.2	Training and development special...
0.2	Materials engineers
0.2	Civil engineers
0.1	Other designers
0.1	Wholesale and retail buyers, exc...
0.1	Physical scientists, all other
0.1	Project management specialists
0.1	Dietetic technicians and ophthal...
0.0	Floral designers
0.0	Farmers, ranchers, and other agr...
0.0	Landscape architects

0.0	Miscellaneous social scientists...
SALES, OFFICE & MANUAL — 36%	
2.8	Secretaries and admin. assistants
1.8	Retail salespersons
1.8	Landscaping and groundskeeping w...
1.6	Exercise trainers
1.4	Supervisors of personal care and...
1.3	Taxi drivers
1.3	Police officers
1.2	Executive secretaries
1.1	Stockers and order fillers
1.0	Other life, physical, and social...
1.0	First-Line supervisors of non-re...
0.9	Title examiners, abstractors, an...
0.8	Personal care and service worker...
0.7	Fast food and counter workers
0.7	First-Line supervisors of retail...
0.7	Couriers and messengers
0.7	Statistical assistants
0.7	Paralegals and legal assistants
0.6	Cashiers
0.6	Recreation workers
0.6	First-line supervisors of landsc...

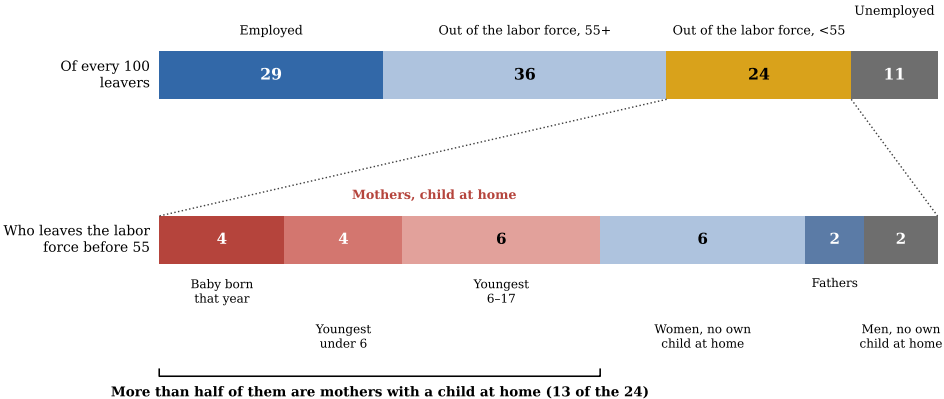
0.6	Legal support workers, all other
0.6	Other healthcare support workers
0.6	Laborers and freight, stock, and...
0.6	Animal caretakers
0.5	Insurance claims and policy proc...
0.5	Personal care aides
0.5	Bill and account collectors
0.4	Waiters and waitresses
0.4	Data entry keyers
0.4	Hairdressers, hairstylists, and...
0.4	Real estate brokers and sales ag...
0.4	Customer service representatives
0.4	First-Line supervisors of office...
0.4	Manicurists and pedicurists
0.4	Other entertainment attendants a...
0.4	Construction laborers
0.4	Office and administrative suppor...
0.3	Cooks
0.3	Bookkeeping, accounting, and aud...
0.3	Miscellaneous agricultural worke...
0.3	Travel agents
0.3	Sewing machine operators
0.3	Bartenders
0.3	Conveyor, dredge, and hoist and...
0.3	Boilermakers

0.3	Billing and posting clerks
0.3	Occupational health and safety s...
0.3	Packaging and filling machine op...
0.2	Laundry and dry-cleaning workers
0.2	Physical therapist assistants an...
0.2	Janitors and building cleaners
0.2	Payroll and timekeeping clerks
0.2	Nursing assistants
0.2	Helpers–production workers
0.1	Hosts and hostesses, restaurant...
0.1	Sales representatives, wholesale...
0.1	Security guards and gambling sur...
0.1	Chefs and head cooks
0.1	Receptionists and information cl...
0.1	Computer, automated teller, and...
0.1	Maintenance and repair workers...
0.1	Office clerks, general
0.0	Stationary engineers and boiler...
0.0	Tellers
0.0	Ushers, lobby attendants, and ti...

Message. 36% of the employed leavers move to occupations where fewer than half of US workers hold a college degree (in red).

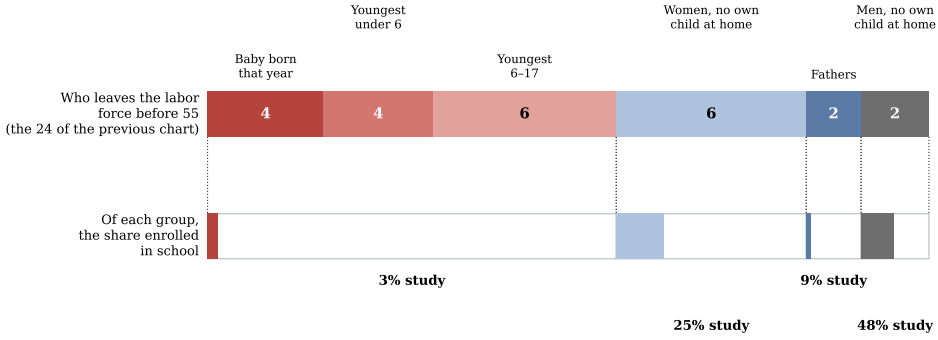
▸ proportions

Who leaves the labor force before 55 (average, 2020–2024)



Message. Early labor-force exit is a family story: more than half of this group are mothers with a child at home.

And the youngest of them leave to study



Overall, 13% of the group is studying; among those under 30, 45%.

Message. Among teachers who leave the labor force before 55, studying is the story of the young and the childless: 45% of those under 30 are enrolled in school, against 3% of the mothers with a child at home.

What this says for policy: three margins, quantified

- **Retirees.** About 3% of teachers retire each year, a third of all exits. Delaying retirement is a *short-term* margin: it extends careers, it does not reintegrate anyone.
- **Switchers.** About 2% of teachers move to another job each year, more than a third of them to jobs that do not require a degree. A real margin, but a moderate one: shortages cannot be closed through retention alone.
- **Early exits.** About 2% of teachers leave the labor force before 55 each year. Mostly mothers at home, for whom *flexibility* may be the missing piece; the young and childless leave to study, and need re-entry paths.

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Correlations: why they leave

Specification

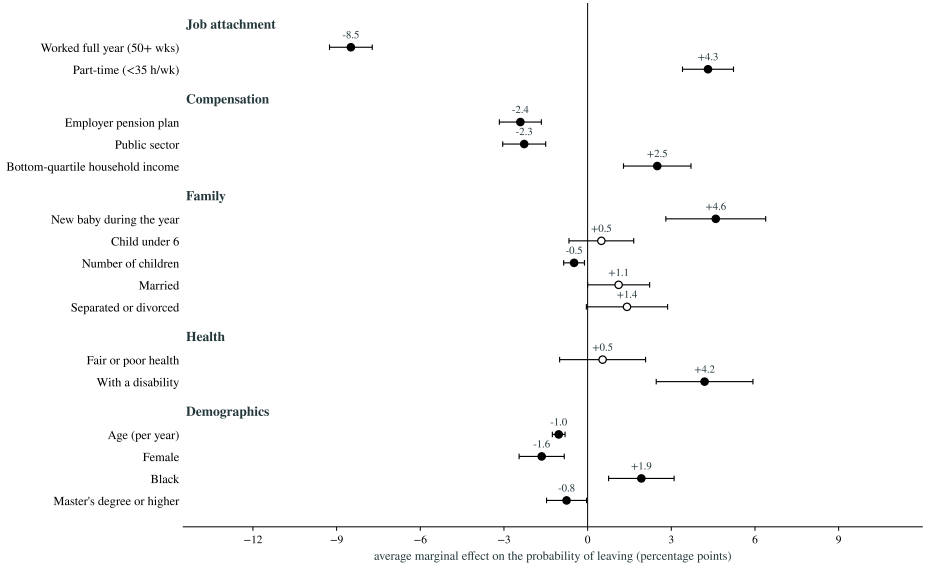
$$y_{it} = \alpha + \underbrace{\mathbf{a}'_{it}\beta_1}_{\text{job}} + \underbrace{\mathbf{c}'_{it}\beta_2}_{\text{pay}} + \underbrace{\mathbf{f}'_{it}\beta_3}_{\text{family}} + \underbrace{\mathbf{h}'_{it}\beta_4}_{\text{health}} + \underbrace{\mathbf{d}'_{it}\beta_5}_{\text{demog.}} + \delta_t + \varepsilon_{it}$$

- y_{it} teacher i left teaching in year t
- $\mathbf{a}_{it}$ job attachment: full-year contract, part-time contract
- $\mathbf{c}_{it}$ pay and security: pension plan, public sector, bottom-quartile household income
- $\mathbf{f}_{it}$ family: new baby, children, marital status
- $\mathbf{h}_{it}$ health: self-reported health, disability
- $\mathbf{d}_{it}$ demographics: age, sex, race, education
- δ_t year fixed effects (state FE variant in the appendix)

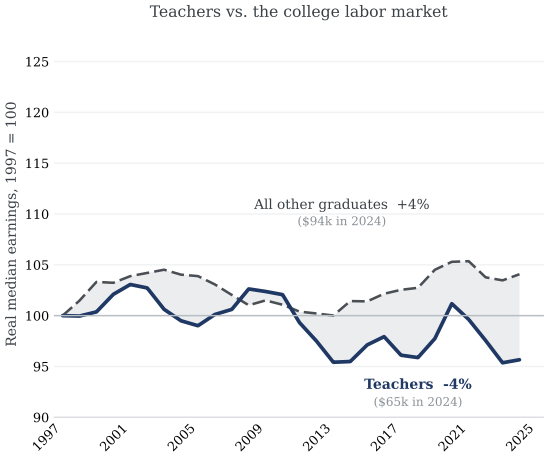
Probit, average marginal effects, robust standard errors. 2018–2024, $n = 18{,}571$.

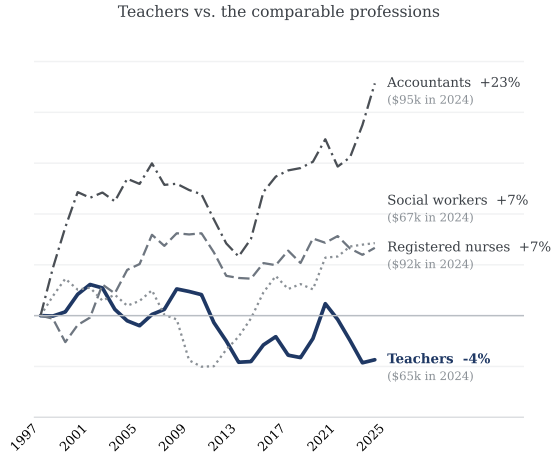
▸ LPM ▸ logit / probit ▸ bounded? ▸ what we observe

What correlates with leaving



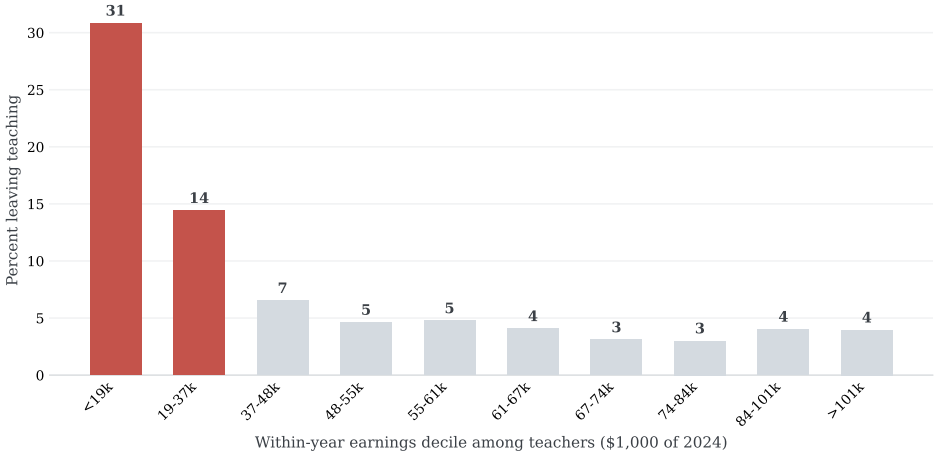
Relative pay since 1997





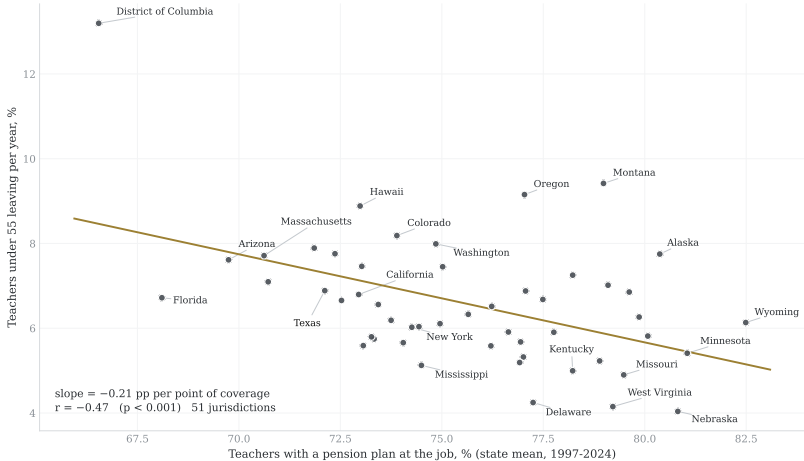
Message. Real teacher pay is 4% *below* its 1997 level while every comparison group gained ground. The gap to other graduates is now $29,000 a year.

Leaving, by position in the teacher pay distribution



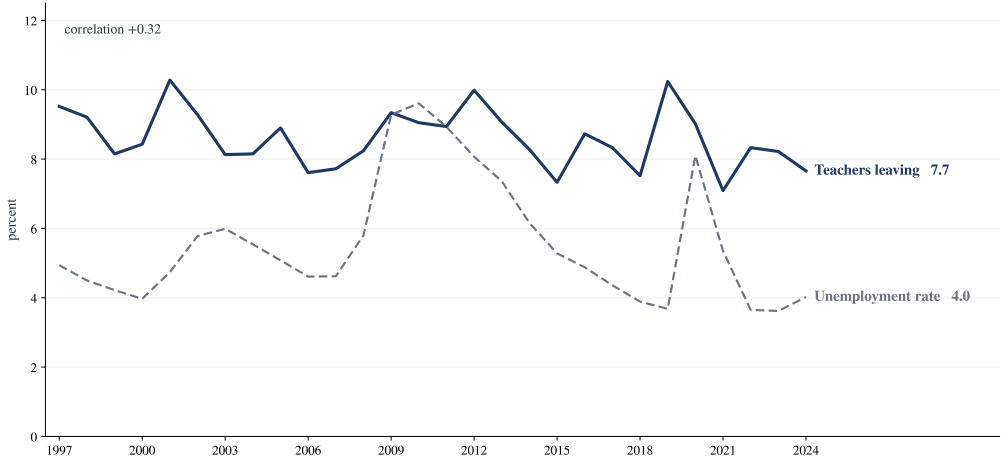
Message. Leaving is concentrated in the bottom two pay deciles, where part-year and part-time contracts live. Above the median, pay barely matters.

Pension coverage and leaving, across states



Message. States where more teachers have a pension plan lose fewer teachers before 55. Economic security retains.

Leaving and the cycle



Message. Leaving rises when unemployment rises (correlation $+0.32$): teachers are pushed out in bad times more than pulled away in good ones.

Conclusions: what we learn

- Teacher attrition is not bad in itself. Much of it is the sign of an active labor market, and it moves with the general cycle. The question for policy is *who* leaves, and why.
- Working conditions weigh heavily: a full-year contract is the single strongest protection against leaving. Job security is fundamental to retain a teacher.
- Not everyone who leaves earns more afterwards, and leaving concentrates at the bottom of the pay scale: the financial lever still has room as a retention incentive.
- A quarter of exits are neither another job nor retirement: care at home, or further study. For that group flexibility is the incentive that can bring them back, and additional qualifications should pay when they return.

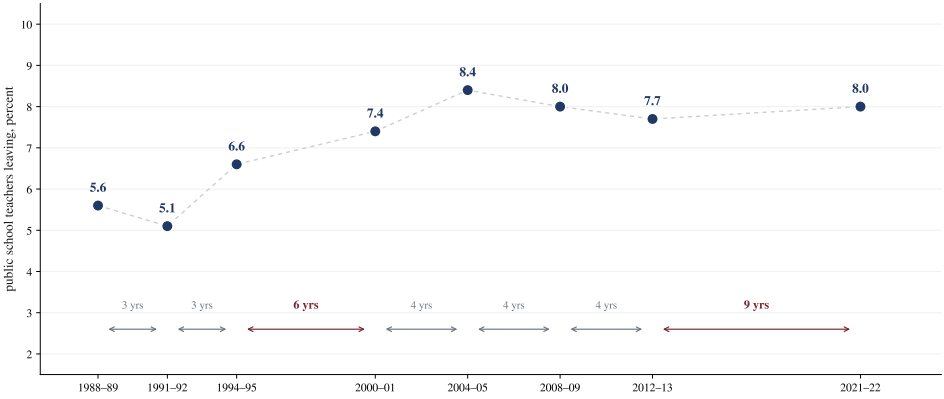
Conclusions: where the policy margins are

- **Short term.** Retirees, about 3% of teachers each year, are a pool that can cover local shortages: experienced people, with positive spillovers on the schools that re-engage them.
- **Medium term.** The young: the probability of leaving is highest at the start of the career. Mentoring and accompaniment belong where the risk is.
- **Across the board.** Economic security (full-year contracts, pension coverage) and flexibility, so that teachers who want to stay, or to return, can; insecurity should not be the reason a position is left.
- **Caveat.** We observe financial and life-cycle motives, not conditions inside schools: these are margins, not the full story.

A

Appendix

Where official numbers exist, and when



The NCES (National Center for Education Statistics) follows up a sample of teachers one year later in the Teacher Follow-up Survey: eight waves since 1988–89, published with a 2–3 year lag, and **nine years** between the last two. The OECD's *Education at a Glance* collects national rates only in its 2021, 2025 and 2026 editions, with no current figure for the United States.

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One survey, three instruments

Instrument	What it asks	What it produces
Basic monthly interview	Labor force status in the reference week, occupation, hours	The official **monthly unemployment rate**; employment and earnings statistics
Topical supplements	School enrollment (Oct.), voting (Nov.), tobacco, computer use	Topic-specific official statistics
March supplement (ASEC)	The *previous calendar year*: longest job, weeks, income, benefits	Official **income, poverty and health-insurance** statistics, and our retrospective measure

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Are interviews monthly? Who is new each month?

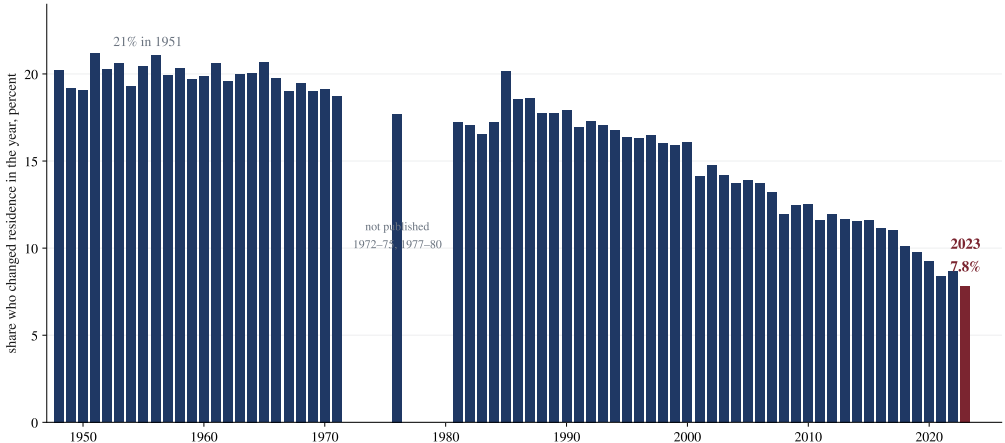
Yes: every month, one eighth of the sample sits in each position of the 4-8-4 rotation. Verified in all 251 monthly files, January 2005 to November 2025:

Month in sample	1	2	3	4	5	6	7	8
Average share (%)	12.4	12.5	12.6	12.5	12.6	12.5	12.4	12.5
Minimum (%)	10.1	11.1	11.5	11.8	12.0	12.0	11.9	12.0
Maximum (%)	13.0	13.0	13.1	13.0	14.9	13.9	13.5	13.2

So each month: $\sim$12.5% new households, $\sim$37.5% in months 2–4, $\sim$50% returning in months 5–8. Month-to-month overlap 75%; year-to-year overlap 50%, by design.

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Internal migration in the United States



In 2023, 7.8% of Americans moved: the loss channel of any address-based panel. Census Bureau, CPS ASEC, historical table A-1.

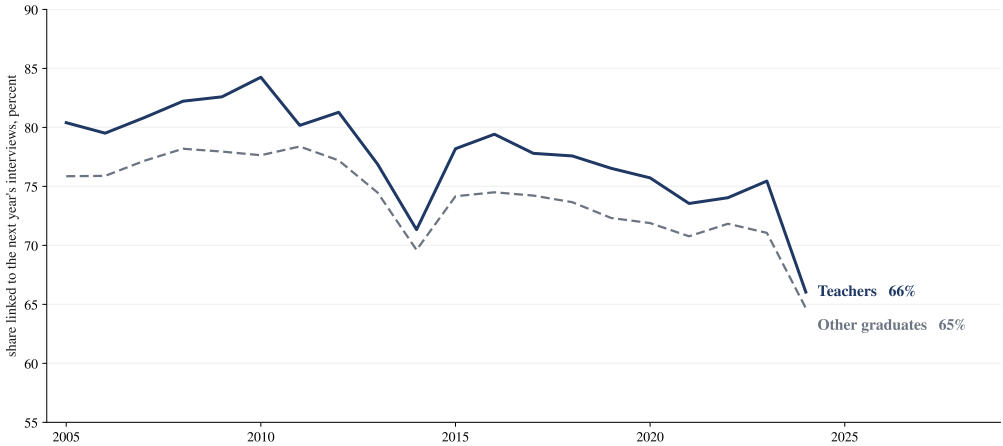
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Is the same person validated year to year?

In the field	In the public files
Roster carried forward by the interviewer	No person identifier at all
Each member keeps a line number	Household id + dwelling id + line number reconstruct it
Occupation asked by dependent interviewing month to month	Validated on sex, race, age advancing 0–2 years
Identity is known	**Identity is inferred**: a mover replaced by someone of the same sex and similar age can slip through

back

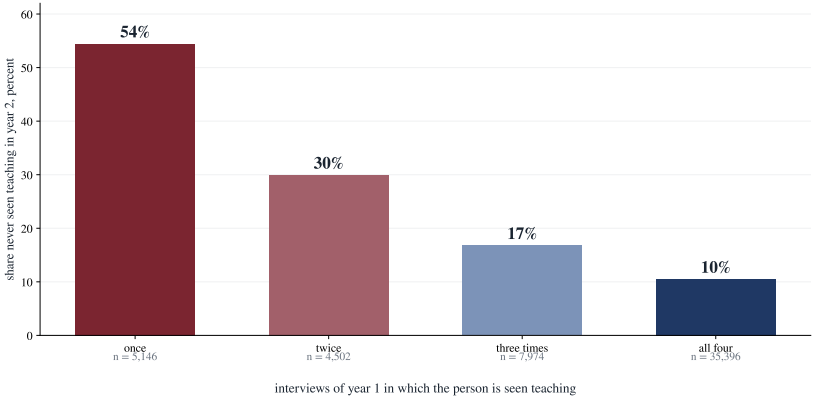
How much is lost in linking, and who



Teachers link *more* than other graduates, not less. But movers leave teaching more, so panel measures are lower bounds on mobility-related exit.

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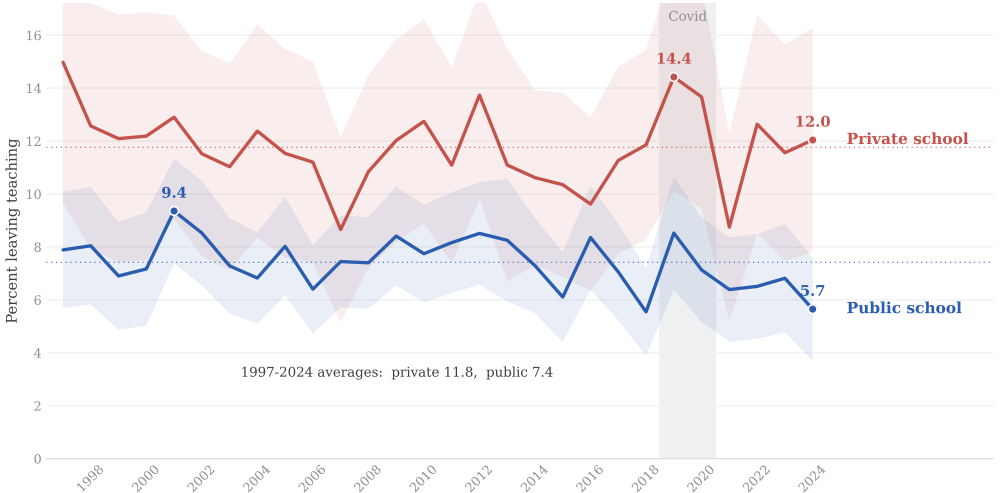
Why the rate falls as the definition tightens



How many times must we see someone teaching in year 1 before calling them a teacher? Seen once: 54% are never seen teaching in their year-2 interviews. Seen in all four: 10%. The stricter the definition, the lower the measured leaving rate. That is the whole ladder of the six definitions.

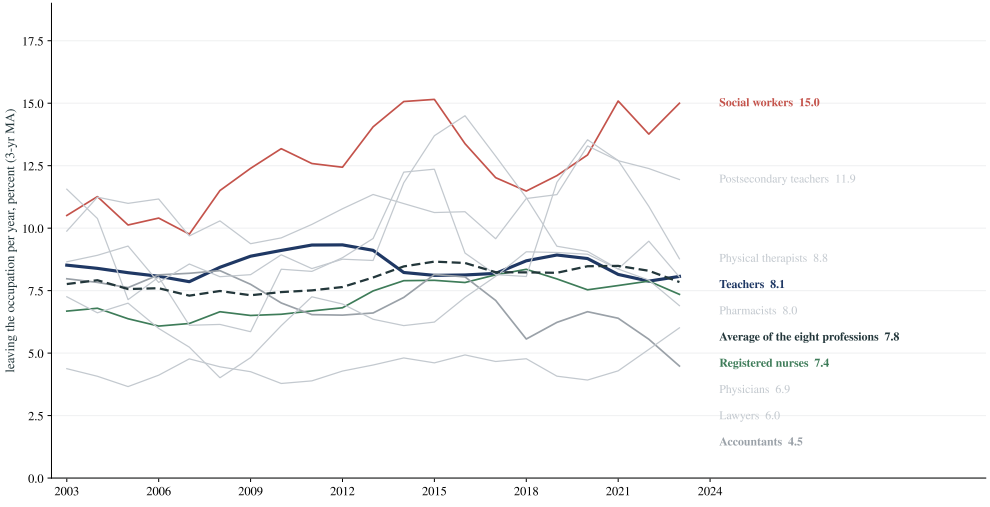
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Public vs private schools



back

More professions, and their average



back

Our destinations vs the NCES follow-up survey

Of every 100 leavers…	CPS 2020–24 avg.	TFS **2021–22**
Working in another job	28.8	57.2
K–12, non-teaching	–	39.3
outside education	–	11.8
Retired / out of LF, 55+	36.4	27.7
Out of the LF, under 55	23.7	10.1
Unemployed	11.1	4.8

TFS: Teacher Follow-up Survey 2021–22, the follow-up of the 2020–21 NTPS. Sources: NCES 2024-039, table A-8, and NTPS estimate tables.

Why the columns differ

1. **Who counts as a teacher.** TFS follows regular classroom teachers of public schools, sampled in schools. We cover public and private, and anyone whose longest job was teaching, including part-year and part-time.
2. **What counts as working.** 39 of the TFS's 57 employed leavers hold *non-teaching K–12 jobs*: they left the classroom, not the school system. Our 28.8 means leaving the teaching occupation.
3. **Who is reachable.** A school-based sample under-represents weakly attached teachers, exactly those who exit to non-participation; a household survey catches them.
4. **The window.** Our column averages five years, 2020–2024; the TFS is the single year 2021–22.

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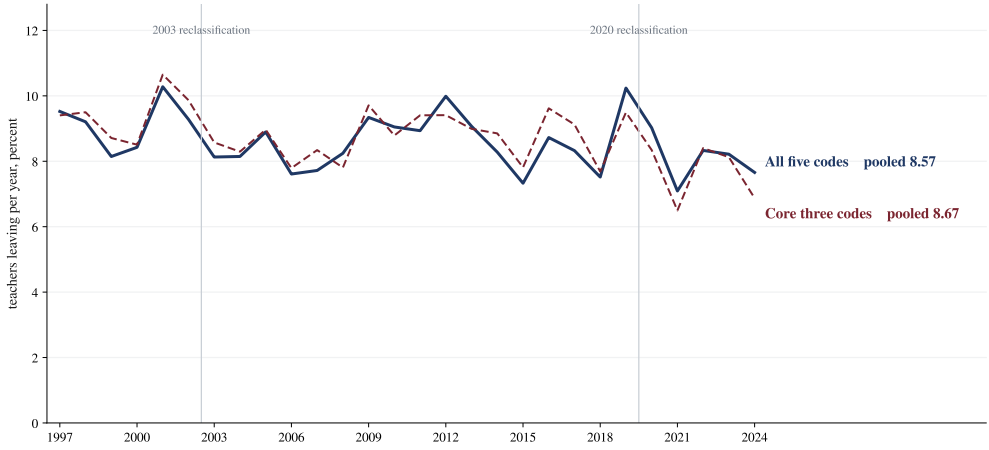
The proportions behind the destinations chart

Of every 100 employed leavers...	share
Education, non-teaching	22.6
Care	8.5
Professional and managerial	32.4
Sales, office and manual	36.5
Memo: occupations where fewer than half of US workers hold a college degree (in red)	36.3

Calendar 2020–2024; 344 employed leavers. The college-degree share of each destination occupation is computed among all employed US workers aged 25–64 in the same ASEC files, 2021–2025. 31.2 of the 36.3 red points sit in the sales, office and manual group; 5.1 in professional occupations.

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Is the teacher group constant across code changes?



Recomputing the whole series on the narrow core moves the pooled rate by 0.1 points; the 2003 and 2020 reclassifications leave no visible break.

back

Linear probability model, fixed-effects variants

	Dependent variable: left teaching (×100)		
	(1)	(2)	(3)
Full-year contract	−11.54***	−11.55***	−11.65***
	(0.67)	(0.66)	(0.67)
Part-time contract	+8.67***	+8.67***	+8.63***
	(1.09)	(1.09)	(1.10)
Employer pension plan	−2.35***	−2.33***	−2.33***
	(0.52)	(0.52)	(0.52)
Public sector	−2.35***	−2.38***	−2.39***
	(0.56)	(0.56)	(0.56)
New baby during the year	+5.72***	+5.63***	+5.71***
	(1.56)	(1.56)	(1.56)
With a disability	+8.85***	+8.94***	+8.94***
	(2.14)	(2.14)	(2.14)
Bottom-quartile household income	+4.17***	+4.13***	+4.01***
	(1.24)	(1.24)	(1.25)
Year fixed effects	No	Yes	Yes
State fixed effects	No	No	Yes
Observations	18,571	18,571	18,571

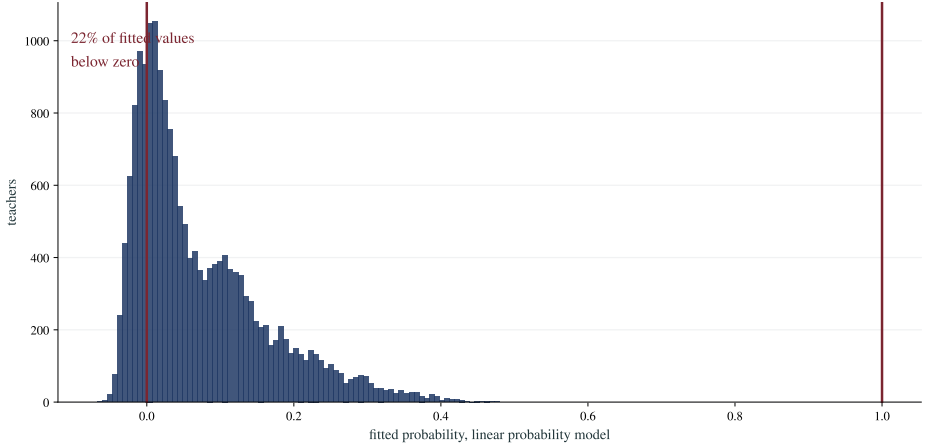
Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Weighted. Full set of controls as in the main specification.

LPM vs logit vs probit

	Average marginal effect on leaving ($\times 100$)		
	LPM (1)	Logit (2)	Probit (3)
Full-year contract	-11.13^{***}	-8.78^{***}	-8.49^{***}
	(0.55)	(0.40)	(0.39)
Part-time contract	$+8.77^{***}$	$+3.86^{***}$	$+4.32^{***}$
	(0.92)	(0.44)	(0.47)
Employer pension plan	-2.55^{***}	-2.38^{***}	-2.41^{***}
	(0.44)	(0.39)	(0.38)
Public sector	-2.17^{***}	-2.24^{***}	-2.27^{***}
	(0.47)	(0.39)	(0.39)
New baby during the year	$+5.65^{***}$	$+4.74^{***}$	$+4.59^{***}$
	(1.33)	(0.90)	(0.91)
With a disability	$+8.31^{***}$	$+3.78^{***}$	$+4.19^{***}$
	(1.84)	(0.80)	(0.88)
Bottom-quartile household income	$+4.67^{***}$	$+2.24^{***}$	$+2.50^{***}$
	(1.10)	(0.58)	(0.62)
Year fixed effects	Yes	Yes	Yes
Observations	18,571	18,571	18,571

Standard errors in parentheses (delta method for the logit and probit marginal effects). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (3) is the

Is the LPM bounded between 0 and 1?



No: 22% of fitted values fall below zero. By that check, the probit is the main specification and the LPM a robustness variant.

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What the CPS observes, of why teachers leave

Reasons in the literature	In the CPS?
Working conditions	no
Organizational support	no
Autonomy and intrinsic motives	no
Financial factors	**yes**
Personal life-cycle motives	**yes**
Sociodemographics	**yes**

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